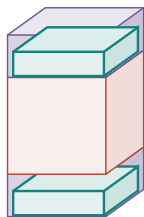


The classes preserving a secant Chern character, and the two conditions they impose, for $n = 4$.

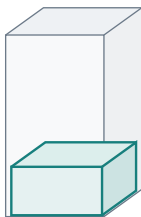
$HT^2(X \times X)$, Künneth stack

$HT^2(X \times \widehat{X})$, dim 120



Φ^{HT}

$HT^1 \otimes HT^1$
kernel 0



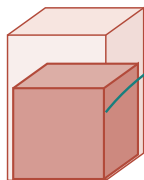
$E = \Phi(F_1 \boxtimes F_2)$: the criterion for E , for $F_1 \boxtimes F_2$, and for F_1 and F_2 are the same, and $F \otimes L$ behaves as F (B -field transform).

$\text{Ann}(\text{ch } F_1) \otimes \frac{1}{n^2}, \frac{1}{n^2} \otimes \text{Ann}(\text{ch } F_2)$

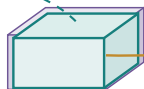
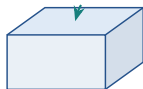
$\text{Ann}(\text{ch } E), 2n^2 = 32$

one factor

(a) $v \lrcorner \text{At}(F) = 0$ on Sym^2 : holds, F moves with (X, Θ)



$z = \frac{d}{2} \pi \lrcorner t^2$



$\text{Ext}^2(F, F)$

$H^1(T_X), 16$
 $\text{Sym}^2: v \lrcorner t = 0, 10$

$H^{0,2}(X), 6$
 $(z, 0, 0)$: never alone

$H^0(\wedge^2 T_X), 6$
all, compensated

(b) $\frac{1}{2} \pi \lrcorner (\text{At}(F)^2 + d t^2) = 0$: open

$HT^2(X)$, dim 28; $\text{Ann}(\text{ch } F)$ has dim $n^2 = 16$

$2n^2$ classes on the eightfold, n^2 on each factor: $n(n+1)/2$ honest deformations of (X, t) and $n(n-1)/2$ Poisson directions with a gerbe compensation. The weakened criterion for $\Phi(F \boxtimes F)$ is (a) and (b), and (b) is $\text{At}(F)^2 = -d t^2 \cdot \text{id}_F$ in $\text{Ext}^2(F, F \otimes \Omega^2)$: the trace free part of the square of the Atiyah class vanishes.